Rheumatology, 2023, **62**, 1552–1558
https://doi.org/10.1093/rheumatology/keac512
Advance access publication 8 September 2022
Original article

British Society for Rheumatology
RHEUMATOLOGY
OXFORD

Clinical science

Survival and prognostic factors from a multicentre large cohort of unselected Italian systemic sclerosis patients

Fabio Cacciapaglia [1,*], Paolo Airò[2], Marco Fornaro [1], Paolo Trerotoli[3], Enrico De Lorenzis [4,5], Addolorata Corrado[6], Maria Grazia Lazzaroni [2], Gerlando Natalello[4], Fabio Montini[1], Alberto Altomare[6], Livio Urso[1], Lucrezia Verardi[4], Silvia Laura Bosello[4], Francesco Paolo Cantatore[5,6], Florenzo Iannone [1]

[1]Department of Emergency and Organ Transplantations, Rheumatology Unit, University and AOUC Policlinico of Bari, Bari, Italy
[2]Rheumatology and Clinical Immunology Unit, ASST Spedali Civili and University, Brescia, Italy
[3]Medical Statistics, Department of Biomedical and Human Oncologic Science, University of Bari, Bari, Italy
[4]Division of Rheumatology, Catholic University of the Sacred Heart, Fondazione Policlinico Universitario Agostino Gemelli IRCCS, Roma, Italy
[5]Ph.D Program in Biomelocular Medicine, Cicle XXXV, University of Verona, Verona, Italy
[6]Rheumatology Clinic, Department of Medical and Surgical Sciences, University of Foggia, Foggia, Italy

*Correspondence to: Fabio Cacciapaglia, Rheumatology Unit, Department of Emergency and Organs Transplantation (DETO), University of Bari, Piazza Giulio Cesare, 11, 70124 Bari, Italy. E-mail: fabio.cacciapaglia79@gmail.com

Abstract

Objectives: Survival and death prognostic factors of SSc patients varied during the past decades. We aimed to update the 5- and 10-year survival rates and identify prognostic factors in a multicentre cohort of Italian SSc patients diagnosed after 2009.

Material and methods: Patients who received a diagnosis of SSc after 1 January 2009 and were longitudinally followed up in four Italian rheumatologic centres were retrospectively assessed up to 31 December 2020. Overall survival of SSc patients was described using the Kaplan–Meier method. Predictors of mortality at 10-year follow-up were assessed by the Cox regression model. A comparison of our cohort with the Italian general population was performed by determining the standardized mortality ratio (SMR).

Results: A total of 912 patients (91.6% females, 20% dcSSc) were included. Overall survival rates at 5 and 10 years were 94.4% and 89.4%, respectively. The SMR was 0.96 (95% CI 0.81, 1.13), like that expected in the Italian general population. Pulmonary arterial hypertension (PAH) and interstitial lung disease (ILD) associated with pulmonary hypertension (PH) significantly reduced survival ($P < 0.0001$). Main death predictors were male gender (HR = 2.76), diffuse cutaneous involvement (HR = 3.14), older age at diagnosis (HR = 1.08), PAH (HR = 3.21), ILD-associated PH (HR = 4.11), comorbidities (HR = 3.53) and glucocorticoid treatment (HR= 2.02).

Conclusions: In the past decade, SSc patients have reached similar mortality of that expected in the Italian general population. Male gender, diffuse cutaneous involvement, comorbidities and PAH with or without ILD represent the main poor prognostic factors.

Keywords: SSc, survival, prognostic factors, pulmonary hypertension, interstitial lung disease

Rheumatology key messages

- The updated 10-year survival of Italian systemic sclerosis patients in the past decade (2009–2019) is approximately 90%.
- Observed SSc standardized mortality ratio was not significantly different from the Italian general population.
- Male gender, diffuse cutaneous involvement, comorbidities and PAH with or without ILD are the main poor prognostic factors.

Introduction

SSc is a multi-organ inflammatory disease characterized by an autoimmune disorder with vasculopathy leading to an excess in collagen and other extracellular matrix proteins deposition, finally resulting in fibrotic and vascular changes in the skin and visceral organs [1, 2]. The disease can involve, besides the skin, the cardiovascular, pulmonary, renal and gastrointestinal systems with potentially progressive damage leading to morbidity, disability and premature death [3, 4].

During the past few decades, different studies evaluated SSc survival with evidence of rising over the years, from <30% of 10-year survival observed in the early 1980s to >80% in the first decade of the 2000s, although a significant improvement in standardized mortality ratio has never been documented [3, 5–7]. In addition, causes of death changed from kidney failure and infective complications as the main causes in the 1990s to cardio-pulmonary involvement reported in the early 2000s in both international and Italian cohorts [8]. Sample

Received: 20 April 2022. **Accepted:** 23 August 2022
© The Author(s) 2022. Published by Oxford University Press on behalf of the British Society for Rheumatology. All rights reserved.
For permissions, please email: journals.permissions@oup.com

size, racial and geographical characteristics, or age and time from disease onset may explain discrepancies in survival rate through different case series [1, 9, 10]. Clinical management can impact SSc survival, and early diagnosis along with new effective treatments availability may affect prognosis [11–13]. Nonetheless, predictors of death may change over time and should be continuously assessed to further improve survival.

Therefore, the present study aimed to update survival and main prognostic factors in a large multicentre cohort of Italian SSc patients diagnosed during the past decade.

Patients and methods

This study evaluated SSc patients longitudinally followed at four Italian rheumatology units with long-term experience in SSc patient management. All consecutive patients fulfilling the classification criteria of 1980 ARA [14] and/or 2013 EULAR/ACR criteria for SSc [15] were considered for the present study. Inclusion criteria for the analysis were: (i) having received a diagnosis of SSc between January 2009 and December 2019 and (ii) at least a 1-year follow-up visit at our centres, or (iii) the availability of data of death, ascertained in any way. Patients without any information after a follow-up visit were considered lost to follow-up and censored at the time of the last available visit. Demographics and disease characteristics were gathered through retrospective clinical chart review, including age at RP onset and age at diagnosis. Patients were classified based on the extent of skin sclerosis as per LeRoy criteria [16]. Interstitial lung disease was defined when bibasilar interlobular and/or interstitial thickening, ground glass opacities and/or honeycombing were detected at high-resolution CT (HRCT) performed according to clinical judgement (crackles at lung's auscultation or restrictive pattern at pulmonary function tests).

In all centres, screening for pulmonary hypertension (PH) was performed using the ESC/ERS guidelines or DETECT algorithm [17] (when they were available) and RHC was performed in patients with an estimated (by echocardiography) increased systolic pulmonary artery pressure or moderate/high probability for PH by the 2013 ESC/ERS guidelines [18]. Precapillary PH was confirmed with right heart catheterization (RHC) when mean pulmonary arterial pressure (mPAP) was $\geq$25 mmHg with a pulmonary wedge pressure (PWP) $\leq$15 mmHg. Pulmonary arterial hypertension (group 1 PH) or interstitial lung disease (ILD)-PH (group 3 PH) was classified based on the absence or presence of forced vital capacity <70% at pulmonary function tests and/or lung HRCT findings, according to ESC/ERS Guidelines [18], respectively. Moreover, we retrospectively applied the updated hemodynamic definitions for PH proposed by the Nizza proceedings [19] mPAP >20 mmHg and pulmonary vascular resistance (PVR) >3 Woods unit.

In addition, the presence during follow-up of digital ulcers, gastrointestinal involvement, categorized as oesophageal, stomach and intestinal symptoms [20], diabetes, systemic hypertension, chronic kidney disease, chronic obstructive pulmonary disease, or other comorbidities identified according to Charlson Comorbidity index, were collected [21].

The presence of serum autoantibodies was tested in all centres for all patients using standard techniques [22]: ANA by indirect immunofluorescence on Hep-2 cell lines (dilution 1:80); anti- ENA, including Anti-topoisomerase-I and ACA antibodies, using chemiluminescence, counterimmunoelectrophoresis or immunoblotting techniques, as availability per centres.

Survival rates were estimated according to follow-up visit information and data on hospital or administrative records. Data were censored at the last available follow-up visit before 31 December 2020, or at the time of death reported on medical archives or administrative sources.

As part of the Systemic sclerosis Progression INvestiGation (SPRING) Italian registry, promoted by the 'Società Italiana di Reumatologia' (SIR) in 2015 as part of SIR-Strategic Projects [4], the study protocol was approved by each ethics committee of the four centres; all patients provided written informed consent to enter in the study with the explicit protection of their identity.

Statistical analysis

Demographics and disease characteristics were evaluated using standard descriptive statistics. Categorical variables were expressed as number or percentage; continuous variables as mean ± standard deviation or median and interquartile range (IQR), as appropriate.

Cumulative survival rates from the diagnosis of SSc were computed by the Kaplan–Meier method [23]. The difference between survival curves of SSc patients without ILD/PH, with ILD alone and belonging to group 1 or 3 PH according to clinical data recorded during follow-up was assessed by the log-rank test, using SPSS v20 (IBM software, New York, NY, USA).

To compare mortality in the SSc cohort to the general Italian population, an indirect standardization method has been applied. The general Italian population data for the years 2009 to the next 10 years was downloaded from the ISTAT website (https://demo.istat.it/ and http://dati.istat.it/Index.aspx). The number of deaths by age class in the Italian population was available for 5 years interval of age, while the total population was assembled accordingly to mortality data. The age classes considered were 20–29, 30–39, 40–49, 50–59, 60–69, 70+. Specific rates by sex and age classes were calculated for the Italian population and then applied to the same strata of the SSc cohort, so to have the expected number of deaths. The ratio between observed and expected deaths is the standardized mortality ratio (SMR), determined with its confidence interval according to the Byar approximation [24]. The SMR and its 95% CI were determined for the whole cohort and a sub-selection with at least 5 years of follow-up since the recruitment date. The SSc cohort has been set up starting with the first enrolled subject with the first diagnosis in 2009 and the last recruited subject in 2019, then followed up to 2020.

To identify potential prognostic factors, the effect of features and manifestations that may impact survival was evaluated with the use of univariate and multivariate Cox regression models, including gender, age at Raynaud onset, age at diagnosis, LeRoy subset, autoantibodies subtypes, ILD, group 1 PH, group 3 PH, other comorbidities, digital ulcers, dilated oesophagus, other gastrointestinal involvement, smoking habit, BMI and treatments. All factors emerging with P-values <0.10 by the single variable analysis were included in the multiple variables model. Then a P-value <0.05 was considered statistically significant.

Results

Overall, 912 patients (91.6% female) with a median (IQR) follow-up from diagnosis of 91.5 (51–120) months were included in the study. Patients' clinical, laboratory and instrumental characteristics are reported in Table 1. Particularly, the median (IQR) age at diagnosis was 52 (40–63) years with a median (IQR) time since RP onset of 24 (0–84.7) months. Diffuse cutaneous involvement was defined in 182 (20%) patients. As shown in Table 1, anti-centromere antibodies (ACA) and anti-topoisomerase-I were detected in 390 (34.7%) and 302 (44.9%) patients, respectively. Of interest, of those 730 patients with limited cutaneous involvement 163 (22.3%) were anti-topoisomerase-I (anti-Scl-70) positive while 382 (52.3%) were ACA positive. Moreover, among other ENA autoantibodies 18 patients were positive for anti-polymerase III, 15 patients were positive for U1-RNP, seven were positive for PM/Scl-100, and three patients resulted positive for Ku. In 28 patients isolated nucleolar at ANA pattern without other ENA antibodies were detectable.

Table 1. Demographics and disease characteristics of SSc population

Variables	*Pts n.912*
Age at RP onset (years), mean (S.D.)	45.2 (16)
Age at diagnosis (years), mean (S.D.)	51 (15.4)
Female, *n* (%)	835 (91.6)
dcSSc, *n* (%)	182 (20)
lcSSc, *n* (%)	730 (80)
ANA positive[a], *n* (%)	843/888 (94.9)
Anti-topoisomerase I (anti-Scl-70)[b], *n* (%)	302/869 (34.7)
ACA[b], *n* (%)	390/869 (44.9)
Other ENA[b], *n* (%)	71/869 (8.2)
ENA negative[b], *n* (%)	106/869 (12.2)
ILD, *n* (%)	472 (51.8)
PAH, *n* (%)	38 (4.2)
Digital ulcers, *n* (%)	395 (43.3)
Oesophageal manifestations, *n* (%)	371 (40.7)
Diabetes, *n* (%)	59 (6.5)
Hypertension, *n* (%)	257 (28.2)
CKD, *n* (%)	61 (6.7)
Gastrointestinal manifestations, *n* (%)	234 (25.7)
COPD, *n* (%)	61 (6.7)
Smoking, *n* (%)	109 (12)
Former smoking, *n* (%)	97 (10.6)
Comorbidities, *n* (%)	406 (44.5)
Endothelin receptor antagonist, *n* (%)	182 (19.9)
PDE5i, *n* (%)	33 (3.6)
Calcium-channel blockers, *n* (%)	437 (47.9)
Biologic agents, *n* (%)	72 (7.9)
RTX	61 (6.7)
TCZ	10 (1.1)
others	2 (0.2)
Any immunosuppressant, *n* (%)	322 (35.3)
AZA	78 (8.6)
MMF	110 (12.1)
MTX	109 (11.9)
Cyc	47 (5.2)
HCQ, *n* (%)	43 (4.7)
Iloprost, *n* (%)	406 (44.5)
Glucocorticoids, *n* (%)	372 (40.8)

Data are expressed as mean (S.D.) or number (percentage).
[a] Missing data for 24 (2.6%) patients.
[b] Missing data for 43 (4.7%) patients.
ACA: anti-centromere antibodies; CKD: chronic kidney disease; COPD: chronic obstructive pulmonary disease; dc: diffuse cutaneous; ILD: interstitial lung disease; ls: limited cutaneous; PAH: pulmonary arterial hypertension; PDE5i: phosphodiesterase-5 inhibitor; RTX: rituximab; TCZ: tocilizumab.

Prevalent non-Raynaud manifestations were: ILD detected in 472 (51.8%), gastrointestinal manifestations in 460 (50.4%) and digital ulcers in 395 (43.3%).

Considering lung involvement, during the follow-up period 574 patients underwent a HRCT scan and in 472 (82.2%) of them findings of ILD were detected, but 115 (24.4%) of them had minimal disease with no progression during follow-up. Among 390 patients positive for ACA, only eight (2%) presented a diffuse cutaneous involvement, but 70/382 (18.3%) patients with limited cutaneous manifestations had ILD with progressive restrictive lung disease.

Regarding gastrointestinal manifestations, 285 (61.9%) patients had dilated oesophagus, 29 (6%) reported only dyspepsia, 33 (7.2%) were constipated and 27 (5.9%) had chronic diarrhoea, while 86 (18.7%) patients presented a combination of these symptoms during the follow-up.

Of note, >43% patients of our cohort presented a history of digital ulcer (DU) during follow-up and were treated with a combination of iloprost, ERA, PDE5i and calcium channel blockers (also as sequential therapy) and almost 10% of them presented refractory DU with the need for adjunctive ulcer dressings, especially during winter.

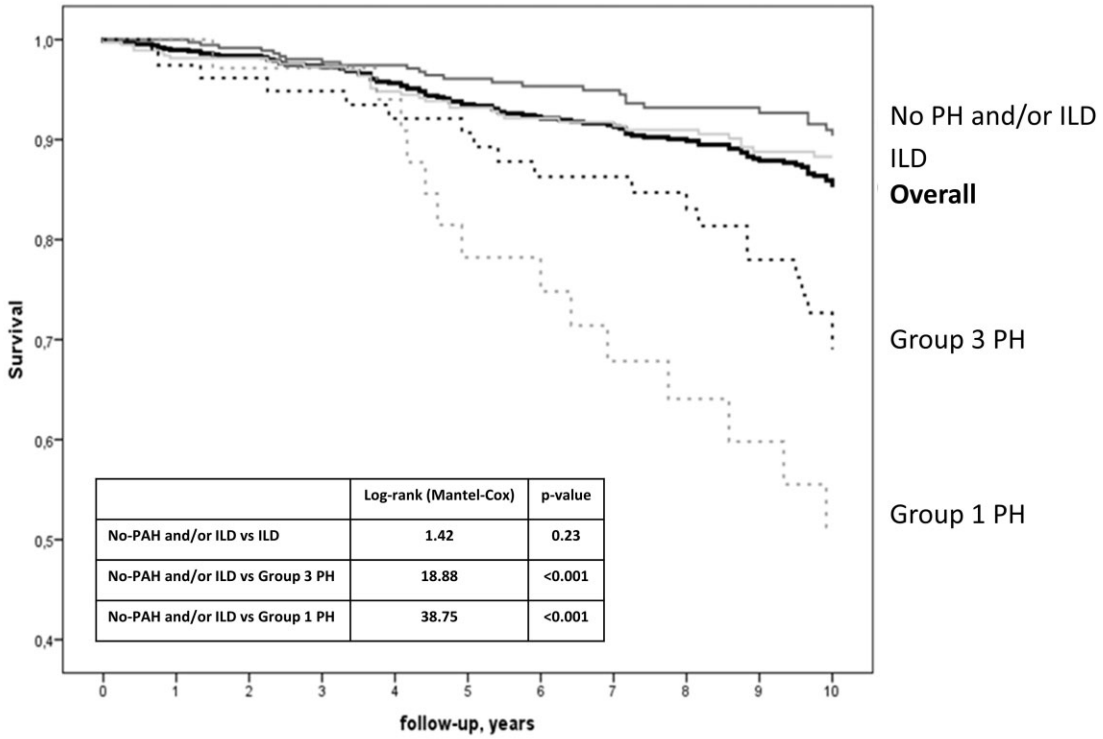
Globally, 162 (17.8%) patients underwent RHC and group 1 PAH was confirmed in 38 (4.2%) patients with RHC. Moreover, 80 (8.8%) patients presented PH associated with ILD (group 3). The retrospective evaluation of our patients according to updated PAH criteria reclassified eight more patients but all of them were already on bosentan for DU at the time of RHC.

Considering treatments, 372 (40.8%), 344 (37.7%) and 72 (7.9%) patients were on glucocorticoids, conventional immunosuppressant and biologic agents, respectively. Of note, 84 (9.2%) patients were on chronic equivalent prednisone dose ≥10 mg/die for joint or lung involvement. All data of specific drugs administered are reported in Table 1.

A total of 890 could be correctly classified by age class and sex with a total of 143 observed deaths, while after standardization expected deaths were 148.6, resulting in an SMR (95% CI) 0.96 (0.81, 1.13), suggesting similar mortality to the Italian population. The procedure was applied to a restricted group with at least 5 years of follow-up that include 608 patients, in which the observed deaths were 95 and expected were 105.6, resulting in an SMR of 0.89 (0.73–1.1), so the mortality could be considered similar to the Italian population for this subgroup too.

Furthermore, we performed a survival analysis dividing SSc patients into four groups according to cardiopulmonary involvement: patients with only ILD, patients with group 1 PH (PAH), patients with group 3 PH (PH+ILD), and patients without PH and/or ILD manifestations. As highlighted from Kaplan–Meier curves, a significantly lower survival in patients with group 1 PH (log-rank = 38.75; 0 < 0.001) and group 3 PH (log-rank = 18.88; $P < 0.001$) compared with patients without PH and/or ILD was detected (Fig. 1). No significant difference in survival was observed between SSc patients without PH and/or ILD manifestations and isolated ILD (log-rank = 1.42; $P = 0.23$).

Subsequently, we evaluated whether the length of the interval between RP onset and SSc diagnosis influenced survival in our cohort. Because the observed median (IQR) interval in our patients was 24 (0–84.7) months, we divided our cohort into two groups accordingly and detected a significantly shorter 10-year survival in those patients with >2 years



	Log-rank (Mantel-Cox)	p-value
No-PAH and/or ILD vs ILD	1.42	0.23
No-PAH and/or ILD vs Group 3 PH	18.88	<0.001
No-PAH and/or ILD vs Group 1 PH	38.75	<0.001

Overall	**885**	**847**	**813**	**764**	**698**	**636**	**575**	**530**	**482**	**437**	**381**
No PH and/or ILD	388	371	354	330	299	269	243	225	199	180	156
ILD	380	364	351	330	301	278	249	231	215	196	172
Group 1 PH	36	35	33	32	30	24	23	19	16	14	12
Group 3 PH	81	77	75	72	68	65	60	55	52	47	41

Figure 1. Kaplan–Meier survival curves in SSc patients. Global and stratified according to presence of isolated interstitial lung disease (ILD), Group 3 pulmonary hypertension (PH) (ILD+PH), Group 1 PH (pulmonary arterial hypertension), or no PH and/or ILD

interval between RP onset and diagnosis (log-rank test: 8.36; $P = 0.004$) (Fig. 2).

Finally, we investigated which factors were associated with survival in our cohort (Table 2). Diffuse cutaneous involvement, interstitial lung disease and anti-topoisomerase-I antibodies having been recognized as relevant prognostic factors in previous survival studies of SSc patients [25, 26] were, however, included in the multivariate analysis even if they did not reach a $P < 0.1$ in our univariate analysis. Cox regression model identified as significant prognostic factors PH diagnosis of group 1 (HR = 3.21, 95% CI: 1.64, 6.29; $P = 0.001$) or group 3 (HR = 4.11, 95% CI: 1.69, 9.96; $P = 0.002$) were associated with an increased risk of mortality. Moreover, male gender (HR = 2.76, 95% CI: 1.31, 5.84; $P = 0.008$), older age at diagnosis (HR = 1.08, 95% CI: 1.05, 1.11; $P = <0.001$), diffuse cutaneous involvement (HR = 3.14, 95% CI: 1.44, 6.83; $P = 0.004$), presence of one or more comorbidities (HR = 3.53, 95% CI: 1.95, 6.37; $P < 0.0001$) and use of glucocorticoids (HR = 2.02, 95% CI: 1.17, 3.49; $P = 0.011$) were other prognostic factors in our SSc cohort. The presence of ILD and anti-topoisomerase-I antibodies didn't reach statistical significance, even after being added to multivariate analysis.

Discussion

This study on a large Italian multicentre cohort of SSc patients has shown a global 10-year survival from diagnosis of ~90%. This finding represents a significant improvement in prognosis compared with previous cohorts over past decades [27–30]. The increase in 5- and 10-year survival observed in this study confirms the trend of previous publications from international studies [31], regardless of the geographical origin of SSc patients. In addition, we observed a 10-year life expectancy of SSc patients as that expected for the Italian general population.

In previous studies, the survival time may have been influenced by diagnosis definition, which varied among different cohorts and over time. Some authors considered survival from RP onset, others from the first non-RP symptom, or when classification criteria were fulfilled. These aspects may impact on the reported survival and should be considered when comparing different SSc studies. In our multicentre study, we assessed survival from diagnosis according to SSc classification criteria available at the patient's assessment time. The last updated 2013 classification criteria for SSc include specific and early manifestations of the disease, and criteria for very early diagnosis of SSc (VEDOSS) allow to intercept and follow up subjects with early manifestations with adequate monitoring of systemic involvement. This condition could have led to a faster diagnosis and therefore survival might appear to be increased because of an earlier diagnosis than in past decades.

Earlier diagnosis can be explained by greater knowledge of clinical frameworks by general practitioners and patient associations, as well as by the systematic use of diagnostic tools and suitable screening programs, such as capillaroscopy and serological analysis. This may lead to a higher referral of SSc patients with milder and/or earlier forms of the disease to specialized rheumatologic centres when compared with previous decades.

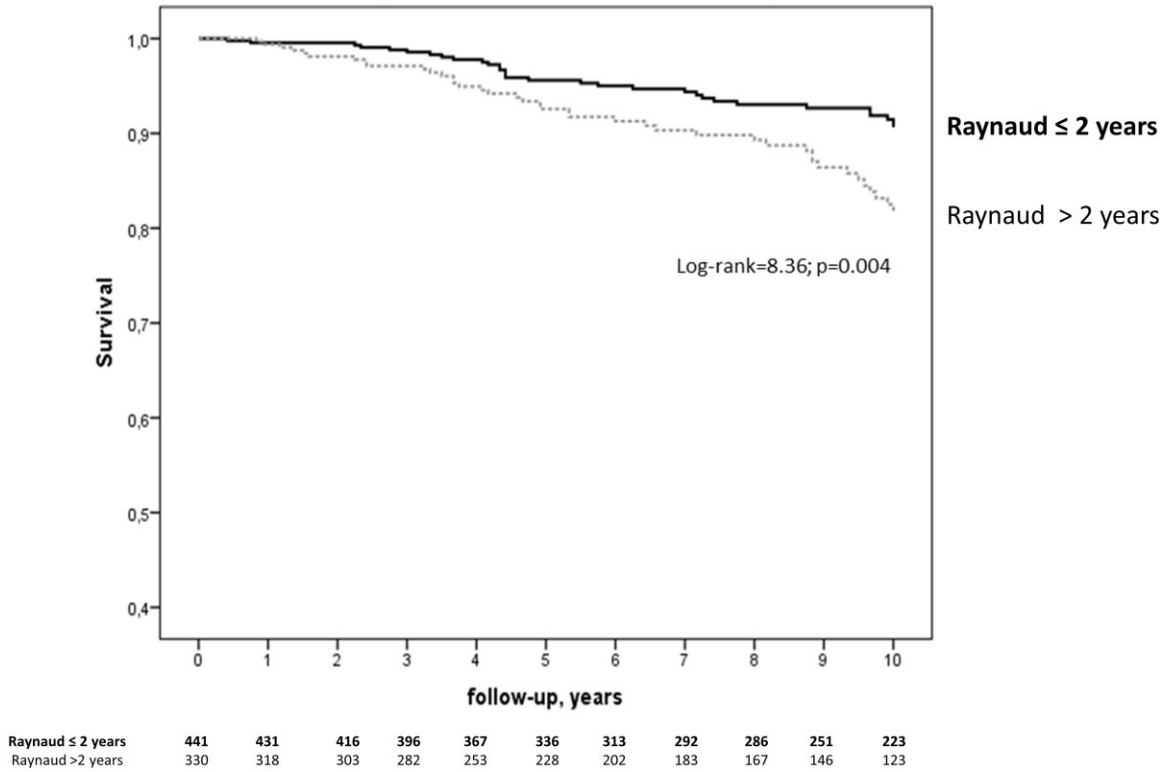


Figure 2. Ten-year survival in SSc cohort according to RP onset. Interval between RP onset and diagnosis: ≤ or > 2 years

Table 2. Univariate and multivariate Cox regression models of prognostic factors in the SSc population

	Univariate			Multivariate		
	OR	95% CI	*P*-value	OR	95% CI	*P*-value
Gender (male *vs* female)	2.22	1.29, 3.78	0.004	2.76	1.31, 5.84	0.008
Age at Raynaud's onset	1.03	1.01, 1.05	<0.001	0.98	0.96, 1.00	0.094
Age at first non-Raynaud symptom	1.08	1.06, 1.10	<0.001	1.08	1.05, 1.11	<0.001
LeRoy subset (dcSSc *vs* lcSSc)	0.98	0.60, 1.62	0.965	3.14	1.44, 6.83	0.004
Antibodies subtypes (Topo-I *vs* others)	0.69	0.43, 1.10	0.120	2.14	0.99, 4.6	0.052
ILD (without PH)	1.27	0.84, 1.91	0.257	1.49	0.73, 3.05	0.266
Group 1 PH	6.17	3.20, 11.87	<0.001	3.21	1.64, 6.29	0.001
Group 3 PH	3.38	1.88, 6.09	<0.001	4.11	1.69, 9.96	0.002
Comorbidities	4.96	3.11, 7.91	<0.001	3.53	1.95, 6.37	<0.001
Digital ulcers	1.14	0.76, 1.71	0.522	–	–	–
Oesophageal manifestations	0.82	0.55, 1.23	0.344	–	–	–
Gasto-intestinal manifestations	1.47	0.77, 2.79	0.242	–	–	–
Smoking	0.89	0.56, 1.4	0.649	–	–	–
BMI	0.97	0.92, 1.03	0.321	–	–	–
Endothelin receptor antagonist	0.79	0.45, 1.37	0.400	–	–	–
Iloprost	0.73	0.49, 1.09	0.127	–	–	–
Calcium-channel blockers	1.13	0.64, 2.01	0.664	–	–	–
Biologic agents	0.29	0.07, 1.19	0.087	0.66	0.15, 3.03	0.596
Any immunosuppressant	1.89	1.08, 3.27	0.024	0.84	0.42, 1.65	0.608
Glucocorticoids	2.29	1.46, 3.57	<0.001	2.02	1.17, 3.49	0.011

dc: diffuse cutaneous involvement; ILD: interstitial lung disease; lc: limited cutaneous involvement; OR: odds ratio; PH: pulmonary hypertension; Topo-I: anti-topoisomerase-I (anti-Scl-70) antibodies.

Indeed, when comparing previous Italian studies that estimated SSc survival, a progressive improvement can be observed. Precisely, Giordano *et al.* in 1986 published their results obtained in a monocentric cohort of 90 scleroderma patients collected between 1965 and 1983 and reported a 72% and 32% rate of 5-year and 10-year survival from SSc diagnosis, respectively [27]. In 2002, Ferri *et al.* presented the first report of a multicentre cohort of 1012 SSc patients from different Italian geographic areas studied from 1955 until 1999. These authors calculated a 5-year survival of 83%, a 10-year survival of 69.2% and 20-year survival of 45.5% from SSc diagnosis [28]. Later, Vettori *et al.* presented a 5-year survival of 94.8% from RP onset between 2000 and 2008 in 251 SSc patients [29]. Finally, with their second

report in 2014, Ferri *et al.* published a 10-year survival of 80.7% from diagnosis in a multicentre cohort of 821 SSc patients recruited between 2000 and 2011 [30]. These studies drew a curve characterized by a progressive improvement in Italian SSc patients' survival during the past decades. It was suggested that this improvement might be at least in part explained by the more frequent recruitment of patients in the early stages of the disease [30]. In agreement with this hypothesis, more recent studies, including ours, recruited larger numbers of patients with favourable prognostic factors (e.g. female, anti-centromere antibody positive, lcSSc) [31]. Our study confirmed this trend with an additional improvement of up to 89.4% 10-year survival. Importantly, in our SSc cohort, we observed that quick diagnosis (<2 years from RP onset) was significantly associated with better survival.

Possible explanations of this finding include both length-time bias, i.e. overestimation of survival duration due to the relative excess of slowly progressing cases that in previous decades would have been diagnosed later, and lead-time bias, i.e. apparently longer survival even in cases when the disease course is the same as in those who were diagnosed later. On the other hand, the longer survival in patients with <2 years from RP onset to SSc diagnosis may have been magnified by the longevity bias: a shorter expectancy of life in those patients with a diagnostic delay at an older age. Despite these limitations, we suggest that quick diagnosis may lead to tight monitoring of early manifestations of the disease and organ involvement, with rapid access to specific effective treatments when needed, and might therefore have a real impact on further survival. The possible role of the availability of new effective immunomodulating and vasoactive treatments for severe heart and lung involvement which may have influenced SSc survival should not be neglected.

In our study, male gender, older age at diagnosis, diffuse cutaneous involvement, use of corticosteroids and pulmonary hypertension were significant negative prognostic factors.

Consistently with our findings, Pokeerbux *et al.* in a recent multicentre study from France identified the same characteristics as being associated with worse prognosis: male sex, advanced age, heart and lung involvement, including both PH and ILD, and diffuse cutaneous involvement [26]. Additionally, the systematic review of the literature and meta-analysis evidenced age at disease onset, male sex, African origin, diffuse cutaneous SSc, anti-topoisomerase-I antibodies, heart and renal involvement, ILD, PH, and malignancy as significantly associated with a worse prognosis, while anti-centromere antibodies positivity were associated to better prognosis [26]. The metanalysis conducted by the same authors in 2019 included 18 studies with 11 719 patients from different geographical areas, and resulted in a pooled SMR of 3.45 (95% CI 3.03, 3.94), while in our SSc cohort we observed no statistically significant differences in SMR compared with the general Italian population.

According to our findings, the recent study from Elhai *et al.* [25] that compared SSc to a national mortality registry in France and EUSTAR cohort data between 2000 and 2011, detected an SMR of 0.8 and recognized male gender, disease duration, diffuse cutaneous, lung and kidney involvement a poor prognostic factors.

Some limitations of our study must be considered: (i) the retrospective collection of data from clinical records with some missing information; (ii) the dissimilar reliability about the onset time of complications in routinary clinical practice and waiting lists of some months for the execution of instrumental examinations; (iii) the heterogeneity in patients' management from different scleroderma units, according to the best care and international recommendations for SSc, although updated during the follow-up period considered in our study; and (iv) the recent consensus-based WSH/HFA definition for primary SSc cardiac involvement [32] did not allow us to retrospectively collect all data about these manifestations.

In conclusion, the management of SSc patients remains challenging as several manifestations impact differently on the prognosis of these patients [33]. The present study highlights that SSc patients are presented a survival without differences from that expected in the general population. The main factors related to the worst prognosis were male gender, older age at diagnosis, diffuse cutaneous involvement, use of corticosteroids and the presence of PH with or without ILD. Physicians who take care of SSc patients should adopt appropriate strategies for screening programs with timely therapeutic intervention to manage these complications in daily clinical practice. Early referral to rheumatologic units with specialized staff dedicated to the management of SSc patients (scleroderma units) is therefore mandatory, especially in the early stages of the disease, identifying a potential 'window of opportunity' as for other immune-mediated inflammatory diseases to further improve prognosis.

Data availability statement

The data supporting this study's findings are available from the corresponding author upon reasonable request. The SSc patients' data are not publicly available due to privacy/ethical restrictions, while data about the Italian general population are available on the ISTAT website at URL https://demo.istat.it/ and http://dati.istat.it/Index.aspx.

Funding

No specific funding was received from any bodies in the public, commercial or not-for-profit sectors to carry out the work described in this article.

Disclosure statement: The authors have declared no conflicts of interest.

References

1. Gabrielli A, Avvedimento EV, Krieg T. Scleroderma. N Engl J Med 2009;360:1989–2003.
2. Varga J, Marangoni RG. Systemic sclerosis in 2016: dermal white adipose tissue implicated in SSc pathogenesis. Nat Rev Rheumatol 2017;13:71–2.
3. Tyndall AJ, Bannert B, Vonk M *et al.* Causes and risk factors for death in systemic sclerosis: a study from the EULAR Scleroderma Trials and Research (EUSTAR) database. Ann Rheum Dis 2010; 69:1809–15.
4. Ferri C, Giuggioli D, Guiducci S *et al.* Systemic sclerosis Progression INvestiGation (SPRING) Italian registry: demographic and clinico-serological features of the scleroderma spectrum. Clin Exp Rheumatol 2020;38(Suppl 125):40–7.
5. Rubio-Rivas M, Royo C, Simeón CP, Corbella X, Fonollosa V. Mortality and survival in systemic sclerosis: systematic review and meta-analysis. Semin Arthritis Rheum 2014;44:208–19.
6. Elhai M, Meune C, Avouac J, Kahan A, Allanore Y. Trends in mortality in patients with systemic sclerosis over 40 years: a systematic review and meta-analysis of cohort studies. Rheumatology 2012; 51:1017–26.

7. Lee YH. Overall and sex- and disease subtype-specific mortality in patients with systemic sclerosis: an updated meta-analysis. Z Rheumatol 2019;78:195–201.
8. Steen VD, Medsger TA. Changes in causes of death in systemic sclerosis, 1972–2002. Ann Rheum Dis 2007;66:940–4.
9. Poudel DR, Derk CT. Mortality and survival in systemic sclerosis: a review of recent literature. Curr Opin Rheumatol 2018;30: 588–93.
10. Komócsi A, Vorobcsuk A, Faludi R *et al.* The impact of cardiopulmonary manifestations on the mortality of SSc: a systematic review and meta-analysis of observational studies. Rheumatology 2012; 51:1027–36.
11. Denton CP, Wells AU, Coghlan JG. Major lung complications of systemic sclerosis. Nat Rev Rheumatol 2018;14:511–27.
12. Moradzadeh M, Aghaei M, Mehrbakhsh Z, Arab-Bafrani Z, Abdollahi N. Efficacy and safety of rituximab therapy in patients with systemic sclerosis disease (SSc): systematic review and meta-analysis. Clin Rheumatol 2021;40:3897–918.
13. Lescoat A, Varga J, Matucci-Cerinic M, Khanna D. New promising drugs for the treatment of systemic sclerosis: pathogenic considerations, enhanced classifications, and personalized medicine. Expert Opin Investig Drugs 2021;30:635–52.
14. Preliminary Criteria for the Classification of Systemic Sclerosis (Scleroderma). Subcommittee for scleroderma criteria of the American Rheumatism Association Diagnostic and Therapeutic Criteria Committee. Arthritis Rheum 1980;23:581–90.
15. van den Hoogen F, Khanna D, Fransen J *et al.* 2013 classification criteria for systemic sclerosis: an American college of rheumatology/European league against rheumatism collaborative initiative. Ann Rheum Dis 2013;72:1747–55.
16. LeRoy EC, Black C, Fleischmajer R *et al.* Scleroderma (systemic sclerosis): classification, subsets and pathogenesis. J Rheumatol 1988;15:202–5.
17. Coghlan JG, Denton CP, Grünig E *et al.* Evidence-based detection of pulmonary arterial hypertension in systemic sclerosis: the DETECT study. Ann Rheum Dis 2014;73:1340–9.
18. Galie N, Humbert M, Vachiery JL *et al.* 2015 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension: the Joint Task Force for the Diagnosis and Treatment of Pulmonary Hypertension of the European Society of Cardiology (ESC) and the European Respiratory Society (ERS): endorsed by: Association for European Paediatric and Congenital Cardiology (AEPC), International Society for Heart and Lung Transplantation (ISHLT). Eur Respir J 2015;46:903–75.
19. Simonneau G, Montani D, Celermajer DS *et al.* Haemodynamic definitions and updated clinical classification of pulmonary hypertension. Eur Respir J 2019;53:1801913.
20. Jaeger VK, Wirz EG, Allanore Y *et al.* Incidences and risk factors of organ manifestations in the early course of systemic sclerosis: a longitudinal EUSTAR study. PLoS One 2016;11:e0163894.
21. Charlson ME, Pompei P, Ales KL, MacKenzie CR. A new method of classifying prognostic comorbidity in longitudinal studies: development and validation. J Chronic Dis 1987;40:373–83.
22. Villalta D, Imbastaro T, Di Giovanni S *et al.* Diagnostic accuracy and predictive value of extended autoantibody profile in systemic sclerosis. Autoimmun Rev 2012;12:114–20120.
23. Fornaro M, Cacciapaglia F, Lopalco G, Venerito V, Iannone F. Predictors of long-term clinical remission in rheumatoid arthritis. Eur J Clin Invest 2021;51:e13363.
24. Rothman KJ, Boice JD Jr. Epidemiologic analysis with a programmable calculator. In: H Austin, ed. With an appendix for the HP-41CV, Chapter 5-Exact testing and estimation, Program 14-Exact confidence limits for an SMR. Boston, MA: Epidemiology Resources, Inc, 1982:30–1.
25. Elhai M, Meune C, Boubaya M *et al.* Mapping and predicting mortality from systemic sclerosis. Ann Rheum Dis 2017;76: 1897–905.
26. Pokeerbux MR, Giovannelli J, Dauchet L *et al.* Survival and prognosis factors in systemic sclerosis: data of a French multicenter cohort, systematic review, and meta-analysis of the literature. Arthritis Res Ther 2019;21:86.
27. Giordano M, Valentini G, Migliaresi S, Picillo U, Vatti M. Different antibody patterns and different prognoses in patients with scleroderma with various extent of skin sclerosis. J Rheumatol 1986;13:911–6.
28. Ferri C, Valentini G, Cozzi F *et al.* Systemic sclerosis: demographic, clinical, and serologic features and survival in 1,012 Italian patients. Medicine 2002;81:139–53.
29. Vettori S, Cuomo G, Abignano G, Iudici M, Valentini G. [Survival and death causes in 251 systemic sclerosis patients from a single Italian center]. Reumatismo 2010;62:202–9.
30. Ferri C, Sebastiani M, Lo Monaco A *et al.* Systemic sclerosis evolution of disease pathomorphosis and survival. Our experience on Italian patients' population and review of the literature. Autoimmun Rev 2014;13:1026–34.
31. Ioannidis JP, Vlachoyiannopoulos PG, Haidich AB *et al.* Mortality in systemic sclerosis: an international meta-analysis of individual patient data. Am J Med 2005;118:2–10.
32. Bruni C, Buch MH, Furst DE *et al.* Primary systemic sclerosis heart involvement: a systematic literature review and preliminary data-driven, consensus-based WSF/HFA definition. J Scleroderma Relat Disord 2022;7:24–32.
33. Giuggioli D, Bruni C, Cacciapaglia F *et al.* Pulmonary arterial hypertension: guidelines and unmet clinical needs. Reumatismo 2021; 72:228–46.